

Build an Autocomplete Search Box

Objective: Implement a basic, interactive autocomplete search feature for Wikipedia page titles.

Time Allotment: You will have 50 minutes for implementation, followed by a short demo and design discussion.

Interface: Your solution can be a terminal-based application or feature a more visual user interface.

Resources: Feel free to leverage any resources you would typically use during development, including Google, AI assistants (like ChatGPT, Gemini), Stack Overflow, and open-source libraries. You may use any programming language.

Process:

- You are encouraged to think aloud as you work.
 - If you prefer to work silently, please check in once you have an initial working version.
- At the 50-minute mark, we will transition to a brief demo of your implementation and a discussion of your design choices.

Data Source:

[wikipedia-latest-titles.txt](#)

Alternative URL if the Notion download doesn't work: <https://charta-public.s3.us-east-2.amazonaws.com/interview/wikipedia-latest-titles.txt>

Evaluation Criteria

The primary goal is to deliver a functional, end-to-end solution. A minimal, working version is preferred over a complex, incomplete one. Your solution will

also be assessed on:

- **Correctness:** How accurately does it meet the specified requirements?
- **Completeness:** How many of the core and additional features are implemented?
- **Code Quality:** Efficiency, readability, maintainability, and structure of your code.

Core Features (Required)

1. **Prefix Matching:** As the user types, suggestions should dynamically appear for Wikipedia titles that begin with the entered query.
2. **Interactivity:** Results must update in real-time as the user types.

Additional Feature Ideas (Optional Enhancements)

Consider implementing any of the following to enhance your solution:

- **Multi-word Support:** Queries like `[char heal]` should match titles such as `[charta health]`.
- **Case Insensitivity:** Queries like `[charta health]` should match titles such as `[Charta Health]`.
- **Result Links:** Display a clickable link for each suggested title. Links are formed by prepending `http://wikipedia.org/wiki/` to an encoded title (spaces replaced by underscores, URL-unsafe characters % encoded).
- **Fuzzy Matching:** For example, `[chrt]` should suggest `[charta]`.
- **Result Ranking:** Implement a scoring function to order suggestions based on relevance.
- **Search History:** When the search box is empty, display a list of recent queries.
- **Snippet Generation:** Scrape or generate a short summary/snippet from Wikipedia pages to display alongside results.
- **Creative Additions:** Any other features you believe would significantly improve the user experience.

